is a linearly dependent set. □ combination of v_1, v_2 and v_4. By Theorem 7, $\{v_1, v_2, v_3, v_4\}$ v_4] (Algebraic property (iii))

Exercise 1.7

In Exercises 1–4, determine if the vectors are linearly independent. Justify each answer.

1. $\begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 7 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ -8 \end{bmatrix}$

Solution: Let, for some $c_1, c_2, c_3 \in R$,

$$c_1 \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 7 \\ 2 \\ -6 \end{bmatrix} + c_3 \begin{bmatrix} 9 \\ 4 \\ -8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

By defn. of matrix eqn. $Ax = b$, we get: $\begin{bmatrix} 5 & 7 & 9 \\ 0 & 2 & 4 \\ 0 & -6 & -8 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

By Theorem 3, this matrix equation has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix} 5 & 7 & 9 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & -6 & -8 & 0 \end{bmatrix}$

$$\begin{bmatrix} 5 & 7 & 9 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & -6 & -8 & 0 \end{bmatrix} \xrightarrow[+3 \times Row2]{Row3 = Row3} \begin{bmatrix} 5 & 7 & 9 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \xrightarrow[Row3 \times (1/4)]{Row3 =} \begin{bmatrix} 5 & 7 & 9 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow[-4 \times Row3]{Row2 = Row2} \begin{bmatrix} 5 & 7 & 9 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[-9 \times Row3]{Row1 = Row1} \begin{bmatrix} 5 & 7 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[\times 1/2]{Row2 = Row2}$$